Tinker Tailor LLM Spy: Reconstructing Deleted Chats from Chromium LevelDB Caches

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Abstract & Background

GenAI portals like ChatGPT, Claude, and Gemini have become standard corporate utilities. Developers routinely copy and paste sensitive source code, internal configurations, and credentials into these clients. When users click 'Delete Chat' or clear histories, they believe it is erased. In reality, it stays on disk. Chromium-based browsers cache conversation telemetry inside client-side IndexedDB databases, backed by Google's LevelDB engine. Since LevelDB uses an append-only LSM-tree, deleted records remain intact in write-ahead logs (.log) and Sorted String Tables (.sst/.ldb) for days or weeks.

The Forensic Challenge

- **File Locks:** Browsers place exclusive OS write locks on LevelDB databases during operation.
- **Triage Barriers:** Traditional LevelDB parsers rely on compiled C++ libraries (plyvel), which cannot be compiled/installed on endpoint machines during live incident response.
- **Data Formats:** Telemetry is packed inside V8 binary serialization arrays and Snappy-compressed blocks.

Chromium LevelDB Architecture

LevelDB stores data in active write-ahead logs (.log) and level-structured tables (.ldb/.sst). We bypass exclusive write locks on running browsers by reading the logs in read-only binary mode ('rb'), allowing us to clone logs in real-time without tripping filesystem blocks.

Reversing V8 ValueSerializer

IndexedDB values are serialized using V8's internal binary ValueSerializer. We implement a zero-dependency varint32 reader and block parser to decode these structures.

- **Varint Decoding:** Rebuilding Google Protocol Buffer varint decoders in Python.
- **String tag 0x22 (OneByte):** Standard ASCII string representation.
- **String tag 0x63 (TwoByte):** UTF-16LE string. The length prefix represents the number of bytes, not characters. Correctly reading exactly length bytes prevents parser offset drift.
- **Object boundaries:** Nesting depth tracking increments on object/array start (0x6f, 0x61) and decrements on end (0x7b) to prevent early parser breakages.

Reassembling Chats & Role Math

V8 encodes array property indices as Smi values (shifted left by 1 bit: actual = Smi >> 1). The role of the speaker (user prompt vs assistant response) is mapped mathematically using index parity:

```
role = 'user' if (index / 2) mod 2 = 1 else 'assistant'
```

Using this, we chronological sort carved records using file modification times (mtime) and database byte offsets.

Incident Scenario 1: Insider Leak

An employee uploads proprietary source code into ChatGPT, then deletes the conversation to bypass logging. Verity bypasses locks, clones logs, decompresses Snappy blocks, decodes V8 serialization, and fully reconstructs the deleted code block in under 5 seconds.

Incident Scenario 2: Token Hijack

Infostealer malware sweeps the browser's IndexedDB path and exfiltrates active session tokens. Because browsers fail to encrypt IndexedDB data (unlike standard cookies/passwords encrypted by DPAPI), the attacker extracts the token from local LevelDB caches, instantly taking over the user's active session without credentials or MFA.

Verity Tool & Mitigations

- **Verity:** Open-source, zero-dependency Python script to audit endpoints and reassemble conversation logs.
- **Integrity:** Seals evidence using HMAC-SHA256 signatures, validated in-browser.
- **Mitigations:** Configure browsers to wipe IndexedDB on exit, enforce BitLocker/FileVault, and flag profile folder accesses via EDR.